

Understanding Statistical Learning Theory (MA6056)

LECTURE SLIDES

July – November 2026

Topic 01 : Linear Regression (Review)

Linear Regression Model

$$y = f(\mathbf{x}) + \epsilon \quad \text{with} \quad f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

- $x_1, \dots, x_p$: input/predictor/feature/regressor
- y : output/target/response
- $\beta_0, \beta_1, \dots, \beta_p$: unknown parameters/coefficients

x_i 's come from different sources:

- quantitative inputs
- transformations of quantitative inputs, such as $\ln x_1, x_2^2$
- interactions between variables, *e.g.*, $x_3 = x_1 x_2$
- numeric or “dummy” coding of the levels of qualitative inputs

Estimators of coefficients

Training data: $\{(x_1, y_1), \dots, (x_N, y_N)\}$

Least squares criterion: Pick β that minimizes the residual sum of squares

$$RSS(\beta) = \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$$

Normal equations:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\beta) = 0$$

LS Estimator: Assuming that $\mathbf{X}$ has full column rank,

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Predicted and fitted values

Predicted value at x_0 :

$$\hat{f}(x_0) = (1 : x_0)^T \hat{\beta}$$

Fitted values:

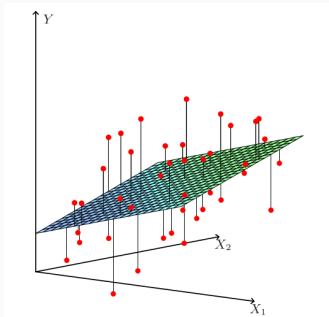
$$\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y$$

Hat matrix:

$$H = X(X^T X)^{-1} X^T$$

Residuals:

$$e = y - \hat{y}$$



Error Assumptions

$$E(\epsilon) = \mathbf{0}$$

$$\text{Var}(\epsilon) = \sigma^2 I_n$$

$$\epsilon \sim N_n(0, \sigma^2 I_n)$$

Properties of LSE of β

$\hat{\beta}$ is a linear function of y

$\hat{\beta}$ is unbiased estimator of β

$$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$$

$\hat{\beta}$ is the BLUE of β .

Estimation of Error Variance (σ^2)

$\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.

Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1}$ is called *Residual variance*. It's square root is called *residual standard error*.

$$\frac{(n-p-1)MS_{Res}}{\sigma^2} \sim \chi^2_{n-p-1}.$$

Here, $SS_{Res} = \sum_{i=1}^n e_i^2 = y^T y - \hat{\beta}_1^T X^T y$

Hypothesis Testing

Test for significance of regression

$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$ ag. $H_1 : \beta_j \neq 0$ for atleast one $j = 1, \dots, p$

Performed using a F statistic

Test for significance of regression coefficients

$H_0 : \beta_j = 0$ ag. $H_1 : \beta_j \neq 0$

Performed using a t statistic

Test of contribution of a subset of the regressors

$H_0 : \beta_2 = 0$ ag. $H_1 : \beta_2 \neq 0$

Performed using a F statistic

Interval estimation

Confidence intervals for β_j :

$$\left[\hat{\beta}_j \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res} C_{jj}} \right]$$

Confidence interval for mean response:

$$\left[\hat{y}_0 \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0} \right]$$

Prediction interval:

$$\left[\hat{y}_0 \mp t_{n-p-1; \frac{\alpha}{2}} \sqrt{MS_{Res} (1 + x_0^T (X' X)^{-1} x_0)} \right]$$

Are the following assumption true?

- Linear relationship
- Error mean zero
- Error variance is constant
- Error are uncorrelated
- Error are normally distributed

Detection of unusual observations: Outliers, influential observations, leverage points

Detection of multicollinearity: Near linear relationship among regressors

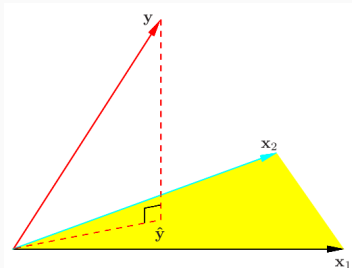
Another graphical interpretation

Denote the column vectors of X by $x_0 \equiv 1, x_1, \dots, x_p$

These vectors span **column space** ($\subset \mathbb{R}^N$) of X

$$\text{Min RSS}(\beta) = \|y - X\beta\|^2$$

Residual vector $y - \hat{y}$ is **orthogonal** to this subspace



This orthogonality is expressed in **normal equation**

$\hat{y}$ is the **orthogonal projection** of y onto this subspace

Hat matrix H computes the orthogonal projection

Suppose we have a univariate model with no intercept:

$$Y = X\beta + \varepsilon$$

The least squares estimate and residuals are:

$$\hat{\beta} = \frac{\sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle}$$

MLR from SLR

Suppose the inputs $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$ are **orthogonal**; that is, $\langle \mathbf{x}_j, \mathbf{x}_k \rangle = 0$ for all $j \neq k$.

Then the multiple least squares estimates $\hat{\beta}_j$ are equal to the univariate estimates:

$$\hat{\beta}_j = \frac{\langle \mathbf{x}_j, \mathbf{y} \rangle}{\langle \mathbf{x}_j, \mathbf{x}_j \rangle}$$

Key Takeaway: When inputs are orthogonal, they have no effect on each other's parameter estimates in the model.

Orthogonalization with an Intercept

Orthogonal inputs rarely occur with observational data. Thus, we orthogonalize them.

Suppose we have an intercept and a single input $\mathbf{x}$. The coefficient of $\mathbf{x}$ is:

$$\hat{\beta}_1 = \frac{\langle \mathbf{x} - \bar{x}\mathbf{1}, \mathbf{y} \rangle}{\langle \mathbf{x} - \bar{x}\mathbf{1}, \mathbf{x} - \bar{x}\mathbf{1} \rangle}$$

where $\bar{x} = \sum_i x_i / N$ and $\mathbf{1} = \mathbf{x}_0$ (vector of N ones).

This is the result of two simple regressions:

1. Regress $\mathbf{x}$ on $\mathbf{1}$ to produce the residual $\mathbf{z} = \mathbf{x} - \bar{x}\mathbf{1}$.
2. Regress $\mathbf{y}$ on the residual $\mathbf{z}$ to give $\hat{\beta}_1$.

Successive Orthogonalization

This recipe generalizes to p inputs:

1. Initialize $\mathbf{z}_0 = \mathbf{x}_0 = \mathbf{1}$.
2. For $j = 1, 2, \dots, p$:
 - Regress $\mathbf{x}_j$ on $\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_{j-1}$ to produce coefficients
$$\hat{\gamma}_{\ell j} = \langle \mathbf{z}_\ell, \mathbf{x}_j \rangle / \langle \mathbf{z}_\ell, \mathbf{z}_\ell \rangle, \quad \ell = 0, \dots, j-1$$
 - and residual vector
$$\mathbf{z}_j = \mathbf{x}_j - \sum_{k=0}^{j-1} \hat{\gamma}_{kj} \mathbf{z}_k.$$
3. Regress y on the residual $\mathbf{z}_p$ to give the estimate $\hat{\beta}_p$.

Result of Successive Orthogonalization

The result of this algorithm is:

$$\hat{\beta}_p = \frac{\langle \mathbf{z}_p, \mathbf{y} \rangle}{\langle \mathbf{z}_p, \mathbf{z}_p \rangle}$$

- Each $\mathbf{x}_j$ is a linear combination of the $\mathbf{z}_k$ ($k \leq j$).
- Since the $\mathbf{z}_j$ are orthogonal, they form a basis for the column space of $\mathbf{X}$.
- More generally, the j -th multiple regression coefficient is the univariate regression coefficient of $\mathbf{y}$ on the residual after regressing $\mathbf{x}_j$ on all other $\mathbf{x}$'s.

Coefficient Interpretation and Variance

The multiple regression coefficient $\hat{\beta}_j$ represents the additional contribution of x_j on y , after x_j has been adjusted for $x_0, x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_p$.

If x_p is highly correlated with other x_k 's, the residual vector z_p will be close to zero, making $\hat{\beta}_p$ very unstable.

The variance is given by:

$$\text{Var}(\hat{\beta}_p) = \frac{\sigma^2}{\langle \mathbf{z}_p, \mathbf{z}_p \rangle} = \frac{\sigma^2}{\|\mathbf{z}_p\|^2}$$

Precision depends on the length of the residual z_p (how much of x_p is unexplained by the other x_k 's).